

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electrical Engineering) SEMESTER - 6 Summer 2026 (Regular)

Course : Bachelor of Technology (Electrical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 6

Subject Code & Name: BTEEPE604A - FLEXIBLE AC TRANSMISSION SYSTEM

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 For a lossless transmission line, maximum power transfer occurs when power angle (δ) is:
A) 45° B) 60° C) 90° D) 180°
- 2 In an HVDC link, extinction angle (γ) is primarily associated with:
A) Rectifier operation B) Inverter operation
C) Transformer operation D) Filter operation
- 3 Which HVDC configuration provides the highest reliability during pole outage?
A) Monopolar link B) Back-to-back link C) Bipolar link D) Homopolar link
- 4 FACTS controllers improve power system performance primarily by controlling:
A) Fuel consumption B) Voltage, impedance and phase angle
C) Generator speed only D) Transformer losses only
- 5 Which FACTS controller belongs to the shunt compensation category?
A) TCSC B) SSSC C) SVC D) IPFC
- 6 Which FACTS controller can provide both capacitive and inductive compensation continuously?
A) Fixed Capacitor B) TSC C) SVC D) Series Reactor
- 7 In a TCR, reactive power is controlled by varying:
A) Supply frequency B) Reactor inductance
C) Firing angle of thyristors D) Transformer ratio
- 8 At $\alpha = 90^\circ$, the current through a TCR is:
A) Minimum B) Maximum C) Zero D) Constant
- 9 TCSC controls power flow by varying:
A) Line resistance B) Generator voltage
C) Effective series reactance D) Load frequency
- 10 Which FACTS controller is a variable impedance series compensator?
A) SVC B) STATCOM C) TCSC D) UPFC

- 11 STATCOM operates as a controllable source of:
 A) Real power only B) Reactive power only
 C) Real and reactive power through energy storage D) Mechanical power
- 12 Which FACTS controller can independently control line voltage, impedance and phase angle simultaneously?
 A) TCSC B) STATCOM C) SSSC D) UPFC
- Q2. Solve the following.
- A) Explain power flow in interconnected AC systems and factors affecting transmission capability. 6
- B) Explain the configuration and components of an HVDC transmission system. 6
- Q3. Solve the following.
- A) Classify FACTS controllers and explain their roles in power systems. 6
- B) Explain the principle and operation of Static VAR Compensator (SVC). 6
- Q4. Solve Any Two of the following.
- A) Explain the importance of reactive power control in transmission systems. 6
- B) Discuss methods of controllable VAR generation. 6
- C) Explain the operation of Thyristor Switched Capacitor (TSC). 6
- Q5. Solve Any Two of the following.
- A) Explain the construction and operation of TSSC. 6
- B) Explain switching converter type series compensator with suitable diagrams. 6
- C) Describe the role of series compensation in power flow control. 6
- Q6. Solve Any Two of the following. <https://www.batuonline.com>
- A) Explain the construction, configuration and operating principle of STATCOM. 6
- B) Compare SVC and STATCOM with suitable characteristics 6
- C) Discuss the configuration and control strategy of Unified Power Flow Controller (UPFC). 6

*** End ***